

Testing Angle Relationships with Dynamic Geometry

Desmos Geometry · Desmos

Year 7	~30 minutes	Mathematics · Geometry & Measurement
--------	-------------	--------------------------------------

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://www.desmos.com/geometry>
Curriculum links: AC9M7M04 · AC9M7M03

Learning intentions

- I can identify corresponding, alternate and co-interior angles formed by a transversal crossing parallel lines.
- I can use a dynamic geometry tool to test whether an angle relationship still holds when the lines are moved.
- I can describe the relationship between a circle's circumference, radius and diameter.

Getting started

Open Desmos Geometry and use the line tool to construct two parallel lines, then draw a third line, a transversal, crossing both. Label the eight angles formed where the transversal crosses each parallel line.

Tasks

1. Use the angle tool to measure two angles that look like a corresponding pair. Record both measurements.

2. Measure four different angle pairs formed by your transversal and record whether each pair is equal or supplementary.

Angle pair type	Angle 1	Angle 2	Equal or supplementary?

3. Drag the transversal to change the angle it makes with the parallel lines. Do your corresponding angles stay equal? What does this tell you about the relationship?

4. Explain, using your measurements, the difference between alternate angles and co-interior angles.

5. Construct a circle using the circle tool. Measure its radius and circumference, then divide the circumference by the diameter. What value do you get, and what does it represent?

6. Repeat the previous task for a circle of a different size. Compare your two results and explain what stays constant no matter the size of the circle.

7. Drag one of your parallel lines so it is no longer parallel to the other. What happens to the angle relationships you found earlier? Explain why.

Extension (optional)

Construct a triangle and measure its three interior angles. Use your understanding of parallel lines and transversals, for example by drawing a line through one vertex parallel to the opposite side, to explain why the three angles always sum to 180 degrees.
